

Internship Progress Report

Chronic Disease Burden & Social Determinants Dashboard

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Organization: Global AI
Period: Week 3, Part-Time Internship

Work Completed

This week had one main deliverable: build an interactive choropleth map showing diabetes prevalence across all U.S. counties, overlaid with Social Vulnerability Index scores, and animatable across the five data years from 2019 to 2023. That sounds straightforward written out like that, but it turned out to be the most technically involved thing I have done so far in this project.

Setting Up the Mapping Environment

The map is built using `folium` with the `TimestampedGeoJson` plugin for the animated layer and `branca` for the colour scales. I had not used any of these before, so the first part of the week was a lot of reading documentation and running small tests to understand how the pieces fit together before writing the actual map code.

The county boundary geometries come from a publicly available GeoJSON file containing all U.S. county polygons keyed by five-digit FIPS code. Getting that file to align correctly with the health panel required careful zero-padding on both sides, which was already handled during the merge step in Week 2, so at least that part was not a surprise.

Animated Diabetes Choropleth

The base layer shows age-adjusted diabetes prevalence for each county, coloured on a yellow-to-red scale clipped to the 2nd and 98th percentiles to prevent extreme outliers from compressing the visible range. Each county is rendered as a separate GeoJSON feature tagged with a `time` property, which is what `TimestampedGeoJson` uses to drive the year slider.

Clicking any county opens a popup showing the county name, state, data year, diabetes and hypertension prevalence, the overall SVI score, its tier label, and all four SVI theme subscores. Getting the popup HTML to render cleanly inside `folium` took more iteration than expected since inline styling inside popup strings is finicky and does not always behave the same way it would in a normal browser context.

SVI Overlay Layer

The second layer is a static SVI choropleth using the 2022 SVI snapshot, coloured on a white-to-dark-blue scale from 0.0 (least vulnerable) to 1.0 (most vulnerable). It is added as a `FeatureGroup` with `show=False` so it is hidden by default and the user can toggle it on via the layer control panel in the top-right corner of the map.

When the SVI overlay is active, hovering over any county shows a tooltip with the overall SVI score, vulnerability tier, all four theme subscores, and the 2023 diabetes and hypertension values. The overlay and the animated diabetes layer can be viewed simultaneously, which is the most useful mode: the orange-to-red diabetes clusters concentrated in the South and parts of Appalachia line up closely with the dark-blue high-vulnerability counties in the SVI layer, which visually reinforces the correlation findings from Week 2.

Output

The finished map is saved as a self-contained HTML file at `results/fig5_diabetes_svi_map.html`. It requires no server and opens directly in a browser.

Difficulties Encountered

- **Learning folium from scratch.** Most of the week was spent understanding how `TimestampedGeoJson`, `FeatureGroup`, and `GeoJson` interact with each other. The plugin documentation is sparse in places, so I had to piece things together from examples and a fair amount of trial and error.
- **Feature count and rendering performance.** The animated layer contains one GeoJSON feature per county per year, which comes out to roughly 15,000 features total. The map is noticeably slow to load in the browser, and stepping through years on the slider has a small lag. I clipped the colour scale to the 2nd and 98th percentiles partly to reduce the number of distinct fill values being computed, but the underlying issue is just the volume of polygon geometry being rendered client-side with no server-side tile support.
- **Missing counties in the boundary file.** 278 counties present in the health panel had no matching geometry in the GeoJSON boundary file. I confirmed these are mostly minor civil divisions and independent cities that are coded differently across datasets, so the mismatch is expected rather than a sign that the join went wrong. Those counties are skipped silently during feature construction.
- **Layer opacity when both layers are active.** When the SVI blue overlay is toggled on at the same time as the diabetes colour layer, the blue fill sits on top and makes the underlying disease gradient harder to read. I set the SVI fill opacity to 0.65 to keep both layers at least partially visible, but it is not a perfect solution. A cleaner fix would be to switch the SVI overlay to a stroke-only style so it does not obscure the fill underneath, which is something I could revisit later if needed.

Planned Work

Next week I plan to start the county-level segmentation step, which means grouping counties into risk tiers based on their combined chronic disease burden and social vulnerability profile. The likely approach is a clustering algorithm such as k-means applied to the merged panel, with the number of clusters chosen using a combination of the elbow method and interpretability. I also want to revisit the binge drinking negative correlation from the Week 2 analysis and check whether it holds within vulnerability tiers once the clusters are defined, since that result is still sitting as an open question.